

Osiris PDB Rev A — Purchasing BOM (links corrected)

Updated from the Oct. 1, 2026 KiCad design and purchasing sheet. Existing priced rows remain legacy budget value

Boards to build5Estimated cost / board

Category	References	Value / Part	Manufacturer
Capacitor	C1	22n	TDK
Capacitor	C2	1U0	TDK
Capacitor	C3, C5, C8, C10	100n	KEMET
Capacitor	C4, C9	10u	TDK
Capacitor	C6	10u	TDK
Capacitor	C7	100n	TDK
Capacitor	C11, C12, C13, C14, C15, C16	1u	TDK
Capacitor	C17	470u 35V HYBRID	Panasonic
Capacitor	C18	1u 50V X7R	KYOCERA AVX
Capacitor	C19	100n 50V X7R	KEMET
Diode / TVS	D1, D4	SMAJ24CA	Bourns
Diode / TVS	D3	STPST10H100SB	STMicroelectronics
Diode / TVS	D5	SMBJ24CA	Littelfuse
Fuse	F1	4A MINI	Littelfuse
Fuse	F2	SCHURTER UHS 100A 3-140-178	SCHURTER
Connector	J1, J3	MKDSP 25/2-15.00 PROVISIONAL	Phoenix Contact
Connector	J2	OSIRIS_I2C	JST

Connector	J4	XT60PW-F	AMASS
MOSFET	Q1, Q2	ISC015N06NM5LF2	Infineon
MOSFET	Q3	BSC070N10NS5	Infineon
Resistor	R1	237k	YAGEO
Resistor	R2	11k8	YAGEO
Resistor	R3	2M0	YAGEO
Resistor	R4	22k	YAGEO
Resistor	R5	5m	Ohmite
Resistor	R6	680k	YAGEO
Resistor	R7	200k	YAGEO
Resistor	R8, R9	10	YAGEO
Resistor	R10, R17 (DNP), R18 (DNP)	4k7	YAGEO
Resistor	R11	1k58	YAGEO
Resistor	R12	1k0	YAGEO
Resistor	R13	54k9	YAGEO
Resistor	R14	22M	YAGEO
Resistor	R15, R16	10k	YAGEO
Resistor	R19	1k	Vishay
Resistor	R20	1	Vishay
Resistor	R21	100	Yageo
IC	U1	LTC4368HMS-1-PBF	Analog Devices

IC	U2	INA228AIDGSR	Texas Instruments
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IC	U3	LT3010EMS8E	Analog Devices
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IC	U4	74AUP1G07GW	Nexperia
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IC	U5	LTC4359IMS8	Analog Devices
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Fuse Holder	F1-holder	MINI fuse holder	Keystone Electronics
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ESTIMATED TOTAL (all unit prices filled; F2 currently stock-blocked)			
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es unless marked otherwise; shipping/tax/assembly excluded.

87.2124

Estimated batch

436.062

Manufacturer P/N	Qty / Board	Board Count (X)	Total Qty	Unit Cost (USD)
C2012C0G2A223J125AC	1	5	5	\$0.30
C2012X7R1H105K125AB	1	5	5	\$0.26
C0603C104K4RACTU	4	5	20	\$0.13
C3216X7R1H106K160AC	2	5	10	\$0.98
C2012X7R1A106K125AC	1	5	5	\$0.33
C1608X7R1H104K080AA	1	5	5	\$0.10
B32529C0105J189	6	5	30	\$0.30
EEH-ZS1V471P	1	5	5	\$2.78
08055C105KAT2A	1	5	5	\$0.27
C0805C104K5RACTU	1	5	5	\$0.18
SMAJ24CA	2	5	10	\$0.53
STPST10H100SB-TR	1	5	5	\$1.44
SMBJ24CA	1	5	5	\$0.57
0297004.WXNV	1	5	5	\$0.39
3-140-178	1	5	5	\$1.07
1932588	2	5	10	\$13.01
BM04B-GHS-TBT(LF)(SN)	1	5	5	\$0.38

XT60PW-F	1	5	5	\$0.65
ISC015N06NM5LF2ATMA1	2	5	10	\$4.07
BSC070N10NS5ATMA1	1	5	5	\$2.24
RC0805FR-07237KL	1	5	5	\$0.10
RC0805FR-0711K8L	1	5	5	\$0.10
RC0805FR-072ML	1	5	5	\$0.10
RC0805FR-0722KL	1	5	5	\$0.10
FC4L64R005FER	1	5	5	\$2.69
RC0805FR-07680KL	1	5	5	\$0.10
RC0805FR-07200KL	1	5	5	\$0.10
RC0805FR-0710RL	2	5	10	\$0.12
RC0805FR-074K7L	1	5	5	\$0.10
RC0805FR-071K58L	1	5	5	\$0.10
RC0805FR-071KL	1	5	5	\$0.10
RC0805FR-0754K9L	1	5	5	\$0.10
RC0805JR-0722ML	1	5	5	\$0.16
RC0805FR-0710KL	2	5	10	\$0.10
CRCW25121K00FKEGHP	1	5	5	\$0.75
CRCW25121R00FKEGHP	1	5	5	\$0.96
RC0805FR-07100RL	1	5	5	\$0.10
LTC4368HMS-1#TRPBF	1	5	5	\$9.32

INA228AIDGSR	1	5	5	\$4.68
LT3010EMS8E#TRPBF	1	5	5	\$6.45
74AUP1G07GW,125	1	5	5	\$0.19
LTC4359IMS8#TRPBF	1	5	5	\$8.72
3568	1	5	5	\$1.31

Availability block	0	Resolved link/
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Cost / Board	Total Cost	Distributor	Buy Link
\$0.30	\$1.50	DigiKey	Supplier page
\$0.26	\$1.30	DigiKey	Supplier page
\$0.52	\$2.60	DigiKey	Supplier page
\$1.96	\$9.80	DigiKey	Supplier page
\$0.33	\$1.65	DigiKey	Supplier page
\$0.10	\$0.50	DigiKey	Supplier page
\$1.78	\$8.91	Mouser	Buy
\$2.78	\$13.90	DigiKey	Buy
\$0.27	\$1.35	DigiKey	Buy
\$0.18	\$0.90	DigiKey	Buy
\$1.06	\$5.30	Mouser	Supplier page
\$1.44	\$7.20	DigiKey	Buy
\$0.57	\$2.85	DigiKey	Buy
\$0.39	\$1.95	Mouser	Supplier page
\$1.07	\$5.37	DigiKey	Buy
\$26.02	\$130.10	Mouser	Buy
\$0.38	\$1.90	Mouser	Supplier page

\$0.65	\$3.23 LCSC	Buy
\$8.14	\$40.70 DigiKey	Supplier page
\$2.24	\$11.20 Mouser	Buy
\$0.10	\$0.50 DigiKey	Supplier page
\$0.10	\$0.50 DigiKey	Supplier page
\$0.10	\$0.50 Mouser	Supplier page
\$0.10	\$0.50 DigiKey	Supplier page
\$2.69	\$13.45 Mouser	Supplier page
\$0.10	\$0.50 Mouser	Supplier page
\$0.10	\$0.50 Mouser	Supplier page
\$0.24	\$1.20 DigiKey	Supplier page
\$0.10	\$0.50 DigiKey	Supplier page
\$0.10	\$0.50 DigiKey	Supplier page
\$0.10	\$0.50 DigiKey	Supplier page
\$0.10	\$0.50 DigiKey	Supplier page
\$0.16	\$0.80 DigiKey	Supplier page
\$0.20	\$1.00 DigiKey	Supplier page
\$0.75	\$3.75 Mouser	Buy
\$0.96	\$4.80 DigiKey	Buy
\$0.10	\$0.50 DigiKey	Buy
\$9.32	\$46.60 Mouser	Buy

\$4.68	\$23.40 DigiKey	Supplier page
\$6.45	\$32.25 DigiKey	Supplier page
\$0.19	\$0.95 DigiKey	Supplier page
\$8.72	\$43.60 Mouser	Buy
\$1.31	\$6.55 DigiKey	Supplier page
\$87.21	\$436.06	

Source URL

<https://www.digikey.com/en/products/detail/tdk-corporation/C2012C0G2A223J125AC/5023347>
<https://www.digikey.com/en/products/detail/tdk/C2012X7R1H105K125AB/2732982>
<https://www.digikey.com/en/products/detail/kemet/C0603C104K4RACTU/411095>
<https://www.digikey.com/en/products/detail/tdk/C3216X7R1H106K160AC/9991386>
<https://www.digikey.com/en/products/detail/tdk-corporation/C2012X7R1A106K125AC/2619198>
<https://www.digikey.com/en/products/detail/tdk/C1608X7R1H104K080AA/513811>
<https://www.mouser.com/ProductDetail/TDK/B32529C0105J189?qs=Xovllr2ZOuY15IPkCXPf9A%3D%3D>
<https://www.digikey.com/en/products/detail/panasonic-industry/EEH-ZS1V471P/9696291>
<https://www.digikey.com/en/products/detail/kyocera-avx/08055C105KAT2A/3881579>
<https://www.digikey.com/en/products/detail/kemet/C0805C104K5RACTU/411169>
<https://www.mouser.com/en/ProductDetail/Bourns/SM-AJ24CA?qs=A3H9pljuJ8RsSQb8zLkILA%3D%3D>
<https://www.digikey.com/en/products/detail/stmicroelectronics/STPST10H100SB-TR/18700338>
<https://www.digikey.com/en/products/detail/littelfuse-inc/SMBJ24CA/285996>
<https://www.mouser.com/ProductDetail/Littelfuse/0297004.WXNV?qs=KOeOmFnJLTo2xybF75gK9A%3D%3D>
<https://www.digikey.com/en/products/detail/schurter-inc/3-140-178/20532481>
<https://www.mouser.com/en/ProductDetail/Phoenix-Contact/1932588?qs=zGjMOL9s0Z%252BnZxcpLOCAEA%3D%3D>
<https://www.mouser.com/ProductDetail/JST-Commercial/BM04B-GHS-TBTLFSN>

https://lcsc.com/product-detail/plug_changzhou-amass-elec-xt60pw-f_C428722.html

<https://www.digikey.com/en/products/detail/infineon-technologies/ISC015N06NM5LF2ATMA1/22157619>

<https://www.mouser.com/en/ProductDetail/Infineon-Technologies/BSC070N10NS5ATMA1?qs=3Bi3m9r5MQYNT0sczKwccQ%3D%3D>

<https://www.digikey.com/en/products/detail/yageo/RC0805FR-07237KL/727754>

<https://www.digikey.com/en/products/detail/yageo/RC0805FR-0711K8L/730505>

<https://www.mouser.com/ProductDetail/YAGEO/RC0805FR-072ML>

<https://www.digikey.com/en/products/detail/yageo/RC0805FR-0722KL/730683>

<https://www.mouser.com/ProductDetail/Ohmite/FC4L64R005FER>

<https://www.mouser.com/ProductDetail/YAGEO/RC0805FR-07680KL>

<https://www.mouser.com/en/ProductDetail/YAGEO/RC0805FR-07200KL>

<https://www.digikey.com/en/products/detail/yageo/RC0805FR-0710RL/730481>

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<https://www.digikey.com/en/products/detail/yageo/RC0805FR-0710KL/727535>

<https://www.mouser.com/en/ProductDetail/Vishay/CRCW25121K00FKEGHP?qs=FlFvgi4rxBxxON%252BaL1o0tw%3D%3D>

<https://www.digikey.com/en/products/detail/vishay-dale/CRCW25121R00FKEGHP/2222582>

<https://www.digikey.com/en/products/detail/yageo/RC0805FR-07100RL/727543>

<https://www.mouser.com/ProductDetail/Analog-Devices/LTC4368HMS-1TRPBF?qs=5aG0NVq1C4xOw3CaDRvs8A%3D%3D>

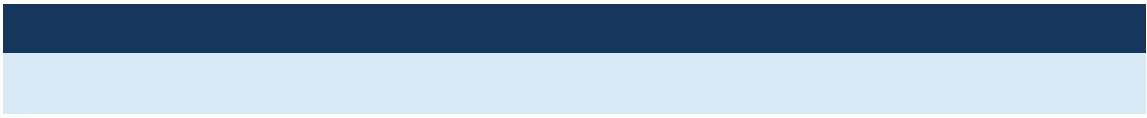
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<https://www.digikey.com/en/products/detail/analog-devices-inc/LT3010EMS8E-TRPBF/960041>

<https://www.digikey.com/en/products/detail/nexperia-usa-inc/74AUP1G07GW-125/1300720>

<https://www.mouser.com/ProductDetail/Analog-Devices/LTC4359IMS8TRPBF?qs=hVkg5c3xu%2Fu7xjpoBvsYg%3D%3D>

<https://www.digikey.com/en/products/detail/keystone-electronics/3568/2137306>



Osiris PDB Rev A — Engineering BOM Detail

Cross-checked against the uploaded current KiCad schematic. C17 and R19 metadata were stale in the purchasing workl

Reference	Quantity	Value	Specification
C1	1	22n	100V C0G 5%
C2	1	1U0	50V X7R
C3	1	100n	16V X7R
C4	1	10u	50V X7R
C5	1	100n	16V X7R
C6	1	10u	10V X7R
C7	1	100n	50V X7R
C8	1	100n	16V X7R
C9	1	10u	50V X7R
C10	1	100n	16V X7R; LT3010 feedback compensation
C11	1	1u	63V film 5%
C12	1	1u	63V film 5%
C13	1	1u	63V film 5%
C14	1	1u	63V film 5%
C15	1	1u	63V film 5%
C16	1	1u	63V film 5%

C17	1	470u 35V HYBRID	Local supplemental ESC capacitor: 470 uF, 35 V hybrid, 11 mOhm ESR at 100 kHz, 4 Arms at 100 kHz/125 C. Preserve the ESC manufacturer-required capacitor at the ESC.
C18	1	1u 50V X7R	50 V X7R ceramic bypass for the ESC branch
C19	1	100n 50V X7R	50 V X7R high-frequency bypass for the ESC branch
D1	1	SMAJ24CA	Bidirectional 400 W TVS; 24 V standoff, 26.7 V min breakdown, 38.9 V clamp
D3	1	STPST10H100SB	100V 10A
D4	1	SMAJ24CA	Bidirectional 400 W TVS; 24 V standoff, 26.7 V min breakdown, 38.9 V clamp
D5	1	SMBJ24CA	Bidirectional 24 V TVS candidate for local ESC connector ringing control; validate pulse energy and clamp voltage on the installed harness.
F1	1	4A MINI	4 A, 32 VDC MINI blade fuse; backup protection for a fault the MOSFETs cannot interrupt
F2	1	SCHURTER UHS 100A 3-140-178	100 A surface-mount high-current fuse candidate for the dedicated ESC branch
H1	0	MountingHole	Mounting Hole without connection
H2	0	MountingHole	Mounting Hole without connection
H3	0	MountingHole	Mounting Hole without connection
H4	0	MountingHole	Mounting Hole without connection
J1	1	MKDSP 25/2-15.00 PROVISIONAL	4S battery input; pin 1 GND, pin 2 VBAT_RAW. Phoenix Contact 1932588 terminal; footprint and assembly require verification.
J2	1	OSIRIS_I2C	JST GH 4-way I2C: 1 NC, 2 SCL, 3 SDA, 4 GND. New Osiris interface and cable continuity require verification.
J3	1	MKDSP 25/2-15.00 PROVISIONAL	Dedicated ESC output after F2; pin 1 GND, pin 2 ESC_VBAT. Phoenix Contact 1932588 terminal; footprint and assembly require verification.
J4	1	XT60PW-F	Avionics raw-4S output to Osiris: pin 1 GND, pin 2 PDB_VOUT. This connector remains on the protected 4 A avionics branch.

Q1	1	ISC015N06NM5LF2	Insulated-Gate Field-Effect Transistor (IGFET), N-Channel, Enhancement, Body Diode, Pin 1 Source, 2 Source, 3 Source, 4 Gate, 5 Drain, 6 Drain, 7 Drain, 8 Drain, 8 Pins
Q2	1	ISC015N06NM5LF2	Insulated-Gate Field-Effect Transistor (IGFET), N-Channel, Enhancement, Body Diode, Pin 1 Source, 2 Source, 3 Source, 4 Gate, 5 Drain, 6 Drain, 7 Drain, 8 Drain, 8 Pins
Q3	1	BSC070N10NS5	100V ideal diode MOSFET
R1	1	237k	Resistor
R2	1	11k8	Resistor
R3	1	2M0	Resistor
R4	1	22k	Series resistor in gate-to-ground timing branch
R5	1	5m	1% 2W
R6	1	680k	Resistor
R7	1	200k	Resistor
R8	1	10	Resistor
R9	1	10	Resistor
R10	1	4k7	Resistor
R11	1	1k58	1% 0.125W; upper feedback resistor
R12	1	1k0	1% 0.125W; lower feedback resistor
R13	1	54k9	Resistor
R14	1	22M	Resistor
R15	1	10k	Resistor
R16	1	10k	Resistor
R17	0	4k7	Optional SCL pull-up; do not populate
R18	0	4k7	Optional SDA pull-up; do not populate
R19	1	1k	Pulse-rated
R20	1	1	Pulse-rated
R21	1	100	1%
TP1	0	TestPoint	test point
TP2	0	TestPoint	test point
TP3	0	TestPoint	test point

TP4	0	TestPoint	test point
TP5	0	TestPoint	test point
TP6	0	TestPoint	test point
TP7	0	TestPoint	test point
			Approved H-temperature-grade LTC4368-1; pin/function-compatible with LTC4368IMS-1 and rated -40 C to 125 C
U1	1	LTC4368HMS-1-PBF	
U2	1	INA228AIDGSR	
U3	1	LT3010EMS8E	Reverse-protected local 3.3 V supply
U4	1	74AUP1G07GW	Single Buffer Gate, Open Drain, Low-Voltage CMOS
U5	1	LTC4359IMS8	Reverse-current blocker
F1-holder	1	MINI fuse holder	Separate purchased holder for F1

book; C18/C19/D5 procurement selections are recorded here.

[illegible]

PDB_Footprints:CP_Elec_Panasonic_G16_10x16.5	Panasonic	EEH-ZS1V471P
Capacitor_SMD:C_0805_2012Metric	KYOCERA AVX	08055C105KAT2A
Capacitor_SMD:C_0805_2012Metric	KEMET	C0805C104K5RACTU
Diode_SMD:D_SMA	Bourns	SMAJ24CA
PDB_Footprints:STPST10H100SB_DPAK	STMicroelectronics	STPST10H100SB-TR
Diode_SMD:D_SMA	Bourns	SMAJ24CA
Diode_SMD:D_SMB	Littelfuse	SMBJ24CA
Fuse:Fuseholder_Blade_Mini_Keystone_3568	Littelfuse	0297004.WXNV
PDB_Footprints:SCHURTER_UHS_8.4x9.4mm_PROVISIONAL	SCHURTER	3-140-178
MountingHole:MountingHole_3.2mm_M3_ISO14580		
MountingHole:MountingHole_3.2mm_M3_ISO14580		
MountingHole:MountingHole_3.2mm_M3_ISO14580		
MountingHole:MountingHole_3.2mm_M3_ISO14580		
PDB_Footprints:MKDSP25_2_15mm_PROVISIONAL	Phoenix Contact	1932588
Connector_JST:JST_GH_BM04B-GHS-TBT_1x04-1MP_P1.25mm_Vertical	JST	BM04B-GHS-TBT(LF)(SN)
PDB_Footprints:MKDSP25_2_15mm_PROVISIONAL	Phoenix Contact	1932588
Connector_AMASS:AMASS_XT60PW-F_1x02_P7.20mm_Horizontal	AMASS	XT60PW-F

PDB_Footprints:ISC015N06NM5LF2_TDS0N8FL	Infineon	ISC015N06NM5LF2ATMA1
PDB_Footprints:ISC015N06NM5LF2_TDS0N8FL	Infineon	ISC015N06NM5LF2ATMA1
PDB_Footprints:BSC070N10NS5_TDS0N8	Infineon	BSC070N10NS5ATMA1
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-07237KL
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-0711K8L
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-072ML
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-0722KL
PDB_Footprints:OH_FC4L64_OHM	Ohmite	FC4L64R005FER
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-07680KL
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-07200KL
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-0710RL
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-0710RL
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-074K7L
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-071K58L
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-071KL
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-0754K9L
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805JR-0722ML
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-0710KL
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-0710KL
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-074K7L
Resistor_SMD:R_0805_2012Metric	YAGEO	RC0805FR-074K7L
Resistor_SMD:R_2512_6332Metric	Vishay	CRCW25121K00FKEGHP
Resistor_SMD:R_2512_6332Metric	Vishay	CRCW25121R00FKEGHP
Resistor_SMD:R_0805_2012Metric	Yageo	RC0805FR-07100RL
TestPoint:TestPoint_Pad_D1.5mm		
TestPoint:TestPoint_Pad_D1.5mm		
TestPoint:TestPoint_Pad_D1.5mm		

TestPoint:TestPoint_Pad_D1.5mm
TestPoint:TestPoint_Pad_D1.5mm
TestPoint:TestPoint_Pad_D1.5mm
TestPoint:TestPoint_Pad_D1.5mm

PDB_Footprints:MS_10-MSOP_ADI	Analog Devices	LTC4368HMS-1#TRPBF
PDB_Footprints:VSSOP10_DGS_TEX	Texas Instruments	INA228AIDGSR

Package_SO:MSOP-8- 1EP_3x3mm_P0.65mm_EP1.68x1.88mm	Analog Devices	LT3010EMS8E#TRPBF
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Package_TO_SOT_SMD:SOT-353_SC-70-5	Nexperia	74AUP1G07GW,125
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Package_SO:MSOP-8_3x3mm_P0.65mm	Analog Devices	LTC4359IMS8#TRPBF
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Fuse:Fuseholder_Blade_Mini_Keystone_3568	Keystone Electronics	3568
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Holder Manufacturer

Holder MPN

Holder Datasheet

Keystone Electronics 3568

[Open datasheet](#)

Selected EEH-ZS1V471P G16; 10 mm diameter, 16.5/16.8 mm class height. Additional ESC-side capacitor and harness transient/ripple tests required.

Procurement selection: 1 uF 50 V X7R 0805, active/stocked; verify effective capacitance under DC bias in final ESC branch.

Procurement selection: 100 nF 50 V X7R 0805, active/stocked; local high-frequency ESC bypass.

Bidirectional 24 V TVS candidate; 38.9 V specified clamp is pulse-condition dependent. Source energy, ringing and downstream peak voltage remain open.

[Open datasheet](#)

Selected power clamp. Connect both anode leads to GND and cathode tab to PDB_VOUT. Pulse/thermal validation required.

[Open datasheet](#)

Selected: local bidirectional clamp from IDEAL_IN to U5 reference IDEAL_VSS.

[Open datasheet](#)

Procurement selection: Littelfuse SMBJ24CA. Retained for local ringing control; validate installed harness pulse energy and clamp behavior.

4 A 32 V MINI; load envelope 2.5 A maximum. Fuse, source and conductor coordination remains a release requirement.

[Open datasheet](#)

PROCUREMENT UPDATE: use 3-140-178 in place of original 3-140-177.

SCHURTER lists identical 100 A / 50 V UHS electrical ratings and package dimensions; order-number difference is packaging. DigiKey offers 3-140-178 cut tape. Engineering release checks still apply: verify 60 A continuous temperature rise, 100 A/30 s time-current behavior, 4S pack fault current, breaking capacity, and heavy-copper land pattern.

[Open datasheet](#)

LAB DESIGN IN PROGRESS: verify exact terminal, footprint, conductor, wave-solder process, and power-path temperature before full-current use.

[Open datasheet](#)

PDB pinout fixed; Osiris R74/R75 12k pull-ups to switched 3.3 V; 100 kHz, ≤ 90 pF total bus capacitance. No power on pin 1. Physical cable mating and new Rev B mapping unverified.

[Open datasheet](#)

LAB DESIGN IN PROGRESS: verify exact terminal, footprint, conductor, wave-solder process, and power-path temperature before full-current use.

[Open datasheet](#)

PDB polarity fixed. Verify physical cable contacts and board-edge clearance; old Osiris J16 numbering is not the new Rev B interface.

[Open datasheet](#)

	Open datasheet
	Open datasheet
Selected: 100 V low-charge MOSFET. Source IDEAL_IN, drain VBAT_FUSED, gate IDEAL_GATE.	Open datasheet
1% 0805 UV divider. Selected for cold-start threshold margin at 12 V; see A-P3 calculation report.	Open datasheet
1% 0805; lower UV divider resistor.	Open datasheet
Package assigned; MPN and power/tolerance validation pending	Open datasheet
U1 GATE directly drives Q1/Q2. R4 is in series with C1 to ground, per LTC4368 Figure 7.	
5 mOhm 1% 2 W four-terminal. Nominal breaker 10 A; normal-condition range 7.92-12.12 A. F1 remains 4 A; conductor/fuse coordination required.	Open datasheet
Package assigned; MPN and power/tolerance validation pending	
Package assigned; MPN and power/tolerance validation pending	
Package assigned; MPN and power/tolerance validation pending	
Package assigned; MPN and power/tolerance validation pending	
Package assigned; MPN and power/tolerance validation pending	
A-P2 U3 feedback network; layout and assembled transient verification required.	Open datasheet
A-P2 U3 feedback network; layout and assembled transient verification required.	Open datasheet
Rev-B candidate; MPN pending distributor check	Open datasheet
22 MOhm 5% gate feedback. Manufacturer order code verified; tolerance included in divider calculation.	Open datasheet
Rev-B candidate; MPN pending distributor check	
Rev-B candidate; MPN pending distributor check	
DNP: host owns pull-up to its switched 3.3 V rail. No local pull-up is allowed with an independently powered host.	
DNP: host owns pull-up to its switched 3.3 V rail. No local pull-up is allowed with an independently powered host.	
Selected: 1k 1% CRCW-HP 2512 reference return. MPN corrected from CRCW25121001FKEGHP to CRCW25121K00FKEGHP. Keep close to U5.	Open datasheet
Selected: 1 ohm 1% CRCW-HP 2512 input damping. Verify pulse energy in layout/bench tests.	Open datasheet
Selected: VBUS filter with C14. Allow 1 ms settling before valid measurements.	Open datasheet
Bare copper test pad; no fitted part required	
Bare copper test pad; no fitted part required	
Bare copper test pad; no fitted part required	

Bare copper test pad; no fitted part required
Bare copper test pad; no fitted part required
Bare copper test pad; no fitted part required
Bare copper test pad; no fitted part required

[Open datasheet](#)

[Open datasheet](#)

A-P2 engineering draft: reverse-input/current-protected LT3010;
R11/R12/C10 set nominal 3.29 V. Verify startup, effective C6 and
assembled rail transients.

[Open datasheet](#)

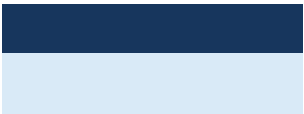
[Open datasheet](#)

Selected: LTC4359 100 V ideal-diode controller; SHDN tied to IN. VSS
return uses R19.

[Open datasheet](#)

Holder qualification pending

[Open datasheet](#)



BOM Role

Schematic component

Schematic component

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DNP

DNP

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Bare PCB feature

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F1 holder; no additional
PCB footprint

Previously Missing / Search-Only Parts — Resolved Purchasing Links

R19 MPN corrected; C17 synchronized to current schematic; C18/C19 exact 0805 parts selected; F2 stock blocker r

Reference	Final Manufacturer P/N	Description
C11, C12, C13, C14, C15, C16	B32529C0105J189	1u
C17	EEH-ZS1V471P	470u 35V HYBRID
C18	08055C105KAT2A	1u 50V X7R
C19	C0805C104K5RACTU	100n 50V X7R
D3	STPST10H100SB-TR	STPST10H100SB
D5	SMBJ24CA	SMBJ24CA
F2	3-140-178	SCHURTER UHS 100A 3-140-178
J1, J3	1932588	MKDSP 25/2-15.00 PROVISIONAL
J4	XT60PW-F	XT60PW-F
Q3	BSC070N10NS5ATMA1	BSC070N10NS5
R19	CRCW25121K00FKEGHP	1k
R20	CRCW25121R00FKEGHP	1
R21	RC0805FR-07100RL	100
U1	LTC4368HMS-1#TRPBF	LTC4368HMS-1-PBF
U5	LTC4359IMS8#TRPBF	LTC4359IMS8

resolved with SCHURTER 3-140-178, the same-spec UHS 100 A / 50 V packaging variant available from DigiKey

Qty for 5 Boards	Supplier	Unit Price (USD)	Extended (USD)	Status
30	Mouser	\$0.30		ORDER
5	DigiKey	\$2.78		ORDER
5	DigiKey	\$0.27		ORDER
5	DigiKey	\$0.18		ORDER
5	DigiKey	\$1.44		ORDER
5	DigiKey	\$0.57		ORDER
5	DigiKey	\$1.07	\$5.37	ORDER
10	Mouser	\$13.01		ORDER
5	LCSC	\$0.65		ORDER
5	Mouser	\$2.24		ORDER
5	Mouser	\$0.75		ORDER
5	DigiKey	\$0.96		ORDER
5	DigiKey	\$0.10		ORDER
5	Mouser	\$9.32		ORDER
5	Mouser	\$8.72		ORDER

/ as cut tape.

Direct Link

<https://www.mouser.com/ProductDetail/TDK/B32529C0105J189?qs=Xovllr2ZOuY15IPkCXPf9A%3D%3D>

<https://www.digikey.com/en/products/detail/panasonic-industry/EEH-ZS1V471P/9696291>

<https://www.digikey.com/en/products/detail/kyocera-avx/08055C105KAT2A/3881579>

<https://www.digikey.com/en/products/detail/kemet/C0805C104K5RACTU/411169>

<https://www.digikey.com/en/products/detail/stmicroelectronics/STPST10H100SB-TR/18700338>

<https://www.digikey.com/en/products/detail/littelfuse-inc/SMBJ24CA/285996>

<https://www.digikey.com/en/products/detail/schurter-inc/3-140-178/20532481>

<https://www.mouser.com/en/ProductDetail/Phoenix-Contact/1932588?qs=zGjMOL9s0Z%252BnZxcpLOCAEA%3D%3D>

https://lcsc.com/product-detail/plug_changzhou-amass-elect60pw-f_C428722.html

<https://www.mouser.com/en/ProductDetail/Infineon-Technologies/BSC070N10NS5ATMA1?qs=3Bi3m9r5MQYNT0sczKwccQ%3D%3D>

<https://www.mouser.com/en/ProductDetail/Vishay/CRCW25121K00FKEGHP?qs=FiFvgi4rxBxxON%252BaL1o0tw%3D%3D>

<https://www.digikey.com/en/products/detail/vishay-dale/CRCW25121R00FKEGHP/2222582>

<https://www.digikey.com/en/products/detail/yageo/RC0805FR-07100RL/727543>

<https://www.mouser.com/ProductDetail/Analog-Devices/LTC4368HMS-1TRPBF?qs=5aG0NVq1C4xOw3CaDRvS8A%3D%3D>

<https://www.mouser.com/ProductDetail/Analog-Devices/LTC4359IMS8TRPBF?qs=hVxkg5c3xu%2Fu7xjpoBvsYg%3D%3D>

Resolution / Note

Exact 1 uF 63 V 5% film capacitor; 30 required for 5 boards. Current Mouser 10+ tier used for budget.

Schematic-selected part. 470 uF, 35 V hybrid, 11 mOhm ESR, 4 A ripple, 10 mm x 16.8 mm SMD can.

Selected procurement part: 1 uF, 50 V, X7R, 0805; active and stocked.

Footprint-compatible with schematic.

Selected procurement part: 0.1 uF, 50 V, X7R, 0805; active and stocked.

Footprint-compatible with schematic.

Exact schematic MPN STPST10H100SB-TR; 100 V, 10 A Schottky in DPAK.

Fixes spreadsheet error where the MPN cell contained a URL. 24 V bidirectional SMBJ TVS, SMB footprint.

Preferred direct drop-in for original 3-140-177. SCHURTER datasheet lists the same 100 A / 50 V ratings, 0.71 mOhm cold resistance, 3200 A²s melting I²t, and same UHS package. 3-140-178 is the 1000-piece reel order number, but DigiKey offers cut tape so only 5 need to be purchased.

Exact Phoenix Contact 1932588 / MKDSP 25/2-15.00. Mouser showed stock for full 10-piece requirement.

Exact AMASS XT60PW-F right-angle female PCB connector; LCSC part C428722.

Exact Infineon BSC070N10NS5ATMA1, 100 V TDSO MOSFET; stocked.

CORRECTED MPN: spreadsheet/schematic had CRCW25121001FKEGHP.

Correct Vishay CRCW-HP 2512 1 kOhm 1% pulse-rated part uses 1K00.

Exact 1 ohm, 1%, 1.5 W, pulse-withstanding 2512 resistor.

Exact 100 ohm, 1%, 0805 Yageo resistor.

Exact LTC4368HMS-1#TRPBF 10-MSOP H-grade surge/UV/OV controller; stocked.

Exact LTC4359IMS8#TRPBF 8-MSOP ideal-diode controller; stocked at Mouser.